

```

(%i1) v1:[1,2];
v1      [1,2]

(%i2) v2:[3,5];
v2      [3,5]

(%i3) A:matrix(v1,v2);
A      
$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$


(%i4) num_of_vector:length(A);
num_of_vector 2

(%i5) r:rank(A);
r      2

(%i6) if r<num_of_vector then
      print("These are linearly dependent")
      else
      print("these are linearly independent");
      these are linearly independent
(%o6) these are linearly independent
(%i7) LC:a·v1+b·v2;
LC      [3 b + a, 5 b + 2 a]

(%i8) v1:[4,-1];
v1      [4, -1]

(%i9) v2:[2,3];
v2      [2,3]

(%i10) B:matrix(v1,v2);
B      
$$\begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix}$$


(%i11) num_of_vector:length(B);
num_of_vector 2

(%i12) r:rank(B);
r      2

(%i13) if r<num_of_vector then
      print("These are linearly dependent")
      else
      print("these are linearly independent");
      these are linearly independent
(%o13) these are linearly independent
(%i14) LC:a·v1+b·v2;
LC      [2 b + 4 a, 3 b - a]

(%i15) v1:[5,-2,4];
v1      [5, -2, 4]

(%i16) v2:[2,-3,5];
v2      [2, -3, 5]

```

(%i17) **v3:[4,5,-7];**

v3 [4, 5, -7]

(%i18) **C:matrix(v1,v2,v3);**

C
$$\begin{bmatrix} 5 & -2 & 4 \\ 2 & -3 & 5 \\ 4 & 5 & -7 \end{bmatrix}$$

(%i19) **num_of_vector:length(C);**

num_of_vector 3

(%i20) **r:rank(C);**

r 2

(%i21) **if r<num_of_vector then**

print("These are linearly dependent")

else

print("these are linearly independent");

These are linearly dependent

(%o21) These are linearly dependent

(%i22) **LC:a·v1+b·v2+c·v3;**

LC [4 c + 2 b + 5 a, 5 c - 3 b - 2 a, -(7 c) + 5 b + 4 a]

(%i23) **v1:[1,2,3];**

v1 [1, 2, 3]

(%i24) **v2:[0,1,4];**

v2 [0, 1, 4]

(%i25) **v3:[2,-1,1];**

v3 [2, -1, 1]

(%i26) **D:matrix(v1,v2,v3);**

D
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$$

(%i27) **num_of_vector:length(D);**

num_of_vector 3

(%i28) **r:rank(D);**

r 3

(%i29) **if r<num_of_vector then**

print("These are linearly dependent")

else

print("these are linearly independent");

these are linearly independent

(%o29) these are linearly independent

(%i30) **LC:a·v1+b·v2+c·v3;**

LC [2 c + a, -c + b + 2 a, c + 4 b + 3 a]

(%i31) **v1:[2,1,-3];**

v1 [2, 1, -3]

```
(%i32) v2:[4,0,5];
```

```
v2      [4,0,5]
```

```
(%i33) v3:[1,-2,1];
```

```
v3      [1,-2,1]
```

```
(%i34) E:matrix(v1,v2,v3);
```

```
E      
$$\begin{bmatrix} 2 & 1 & -3 \\ 4 & 0 & 5 \\ 1 & -2 & 1 \end{bmatrix}$$

```

```
(%i35) num_of_vector:length(E);
```

```
num_of_vector 3
```

```
(%i36) r:rank(E);
```

```
r           3
```

```
(%i37) if r<num_of_vector then
```

```
    print("These are linearly dependent")
```

```
else
```

```
    print("these are linearly independent");
```

```
these are linearly independent
```

```
(%o37) these are linearly independent
```

```
(%i38) LC:a·v1+b·v2+c·v3;
```

```
LC      [c+4 b+2 a,a-2 c,c+5 b-3 a]
```